Department of Electrical and Electronic Engineering

Jamalpur Science & Technology University



Experiment No:02

Experiment Name:Study of Open Circuit Characteristics of Separately Excited DC Generator.

Course Name: Electrical Machine I Sessional

Course Code: EEE 2112

Submitted by :

MD. Maruf Hossen

ID: 25010627

Session: 2024-25

Department of Electrical and Electronic Engineering

Jamalpur Science & Technology University

Submitted to :

MD. Mahfuzul Haque

Assistant Professor

Department of Electrical and Electronic Engineering

Jamalpur Science & Technology University

Experiment No: 02

1. Experiment Name

Study of Open Circuit Characteristics of Separately Excited DC Generator.

2. Objective

1. To study the open circuit characteristics of a separately excited DC generator.
2. To determine the relationship between generated EMF and field current.
3. To plot the Open Circuit Characteristic (OCC) curve of the DC generator.

3. Apparatus

- Separately excited DC generator
- DC motor/prime mover
- DC power supply
- Rheostat
- Ammeter
- Voltmeter
- Tachometer
- Connecting wires
- Switch

4. Theory

A separately excited DC generator is a generator in which the field winding is supplied from an external DC source.

According to Faraday's law of electromagnetic induction, when the armature conductors rotate and cut the magnetic flux, an EMF is induced in the armature.

The generated EMF is given by,

$$E_g = \frac{P\Phi ZN}{60A} \tag{1}$$

where,

- E_g = Generated EMF (V)
- P = Number of poles
- Φ = Flux per pole (Wb)
- Z = Total number of armature conductors
- N = Speed of armature (rpm)
- A = Number of parallel paths

For constant speed,

$$E_g \propto \Phi \tag{2}$$

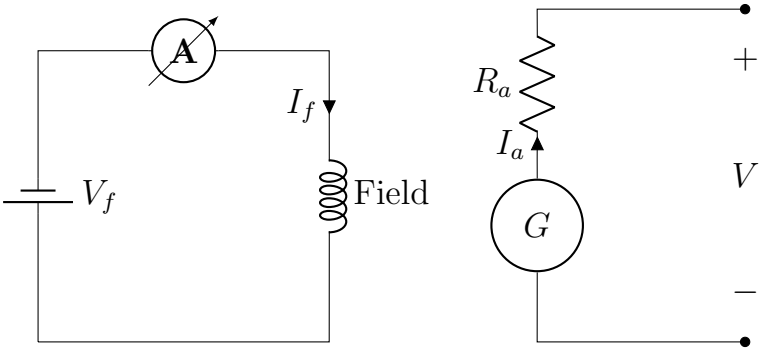
Since the flux Φ depends on field current I_F,

$$E_g \propto I_F \tag{3}$$

at the unsaturated region.

The graph between generated EMF E_g and field current I_F is called the **Open Circuit Characteristic (OCC)** of a DC generator.

5. Circuit Diagram



6. Procedure

1. The circuit was connected according to the given circuit diagram.
2. The field winding was connected to a separate DC supply through a rheostat.
3. The generator was driven by the prime mover.

4. The generator speed was adjusted to the rated value and was kept approximately constant.

5. The field current was initially set to a minimum value.

6. The field current was gradually increased using the rheostat.

7. The corresponding generated EMF was measured using a voltmeter.

8. The readings of field current and generated EMF were recorded in the observation table.

9. The field current was increased up to the required maximum value.

10. The obtained readings were plotted on a graph with I_F on the X-axis and E_g on the Y-axis.

7. Observation Table

Table 1: Observation of Open Circuit Characteristics

Sl. No.	E_g **(V)**	I_F **(mA)**
1	4.53	0.004
2	25.60	11.40
3	62.34	27.11
4	100.22	45.56
5	126.14	60.77
6	154.45	78.57
7	168.64	90.01
8	181.90	106.05
9	195.90	130.15
10	201.12	147.23
11	205.75	156.56
12	210.01	176.77

8. Graph

The Open Circuit Characteristic (OCC) curve was plotted between generated EMF E_g and field current I_F.

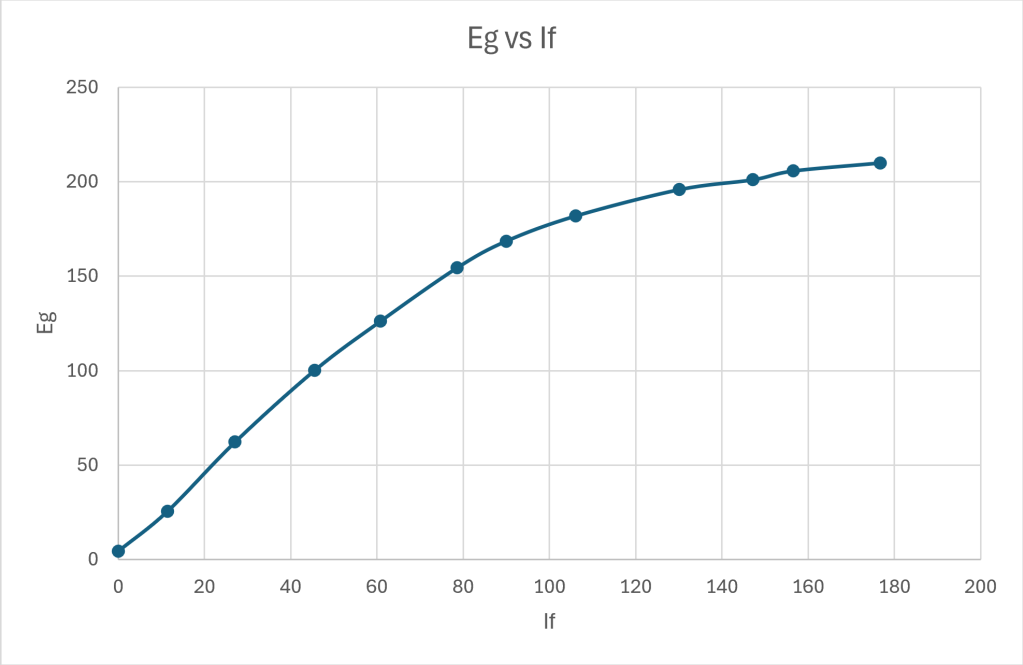


Figure 1: OCC curve of separately excited DC generator

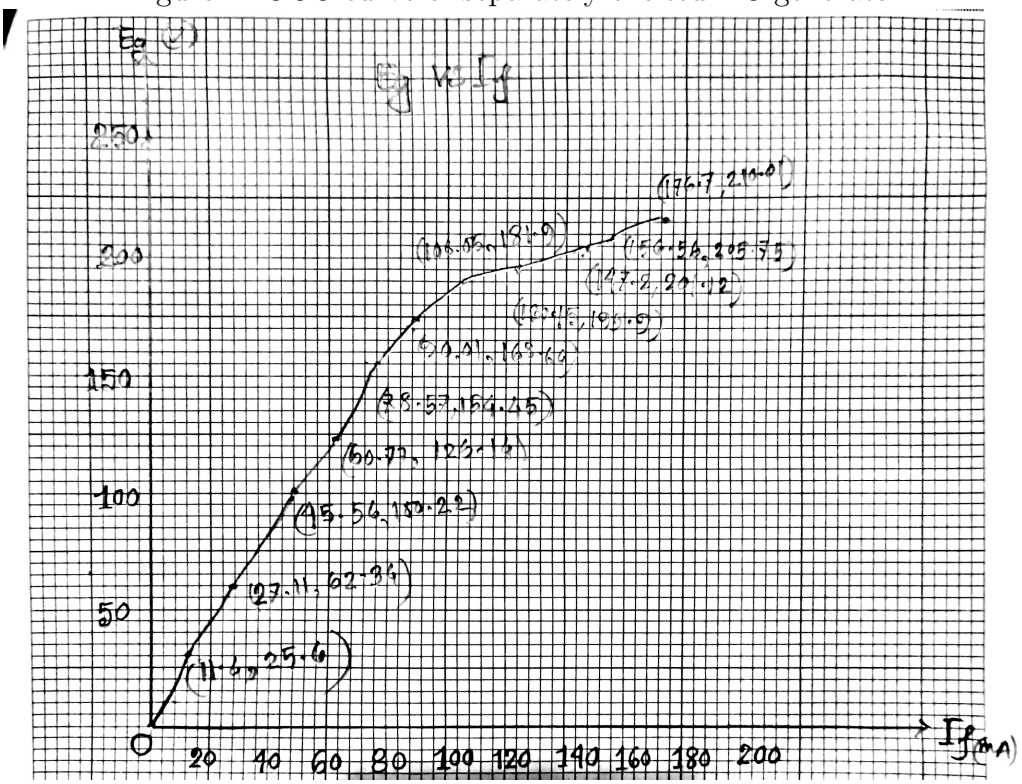


Figure 2: OCC curve of separately excited DC generator-hand drawn

9. Result and Decisions

From the open-circuit characteristics of the separately excited DC generator, it was observed that a residual voltage of 4.53 V was generated at $I_f = 0.004\,A$, confirming the presence of residual magnetism. The generated voltage increased almost linearly with field current up to about $I_f = 80\,A$, reaching 154.45 V. The curve then started bending between $I_f \approx 80 - 100\,A$, indicating the knee point and the beginning of magnetic saturation. Beyond $I_f > 100\,A$, further increase in field current produced only a small increase in generated voltage, reaching approximately 210.01 V at $I_f = 176.77\,A$.

Therefore, the generator should preferably be operated near the knee region, approximately at $E_g = 160-180\,V$ and $I_f = 80-100\,A$, to avoid excessive field copper losses and heating. The observed residual voltage also indicates that the generator has the capability of self-excitation. For practical operation, a nominal output voltage of approximately 180-200 V may be considered for suitable voltage stability and efficient operation.

10. Precautions

1. The generator speed was kept approximately constant throughout the experiment.
2. The field current was increased gradually.
3. The field current was not allowed to exceed the rated value.
4. The polarity of the DC supply was checked before making connections.
5. Proper meter ranges were selected before taking readings.
6. Loose connections were avoided.
7. The generator was switched off after completing the experiment.

11. Conclusion

The OCC of the separately excited DC generator was obtained successfully. It was observed that the generated EMF increased with an increase in field current, while the rate of increase gradually decreased at higher field currents due to magnetic saturation.